

Qatar-1b

R-1717

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_160526 · created 2026-08-02T16:09:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457534.84691 +/- 0.00067 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.013
Transit depth (Rp/Rs)^2	1.97 +/- 0.36 [%]
Orbital Inclination (inc)	87.0 +/- 1.61
Airmass coefficient 1 (a1)	1.043 +/- 0.015
Airmass coefficient 2 (a2)	-0.0276 +/- 0.0068
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	#2 - [312, 52]
Optimal Aperture	2.8
Optimal Annulus	6.94
Transit Duration (day)	0.0785 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.14 $\pm$ 0.013 vs 0.1463 (deviation 0.32 σ)
Duration fit vs published	0.0785 d vs 0.0692 d (deviation 1.46 σ)
Mid-time O-C	2.0 min
Depth SNR	7.1
Residual scatter	1.47 %

Light curve

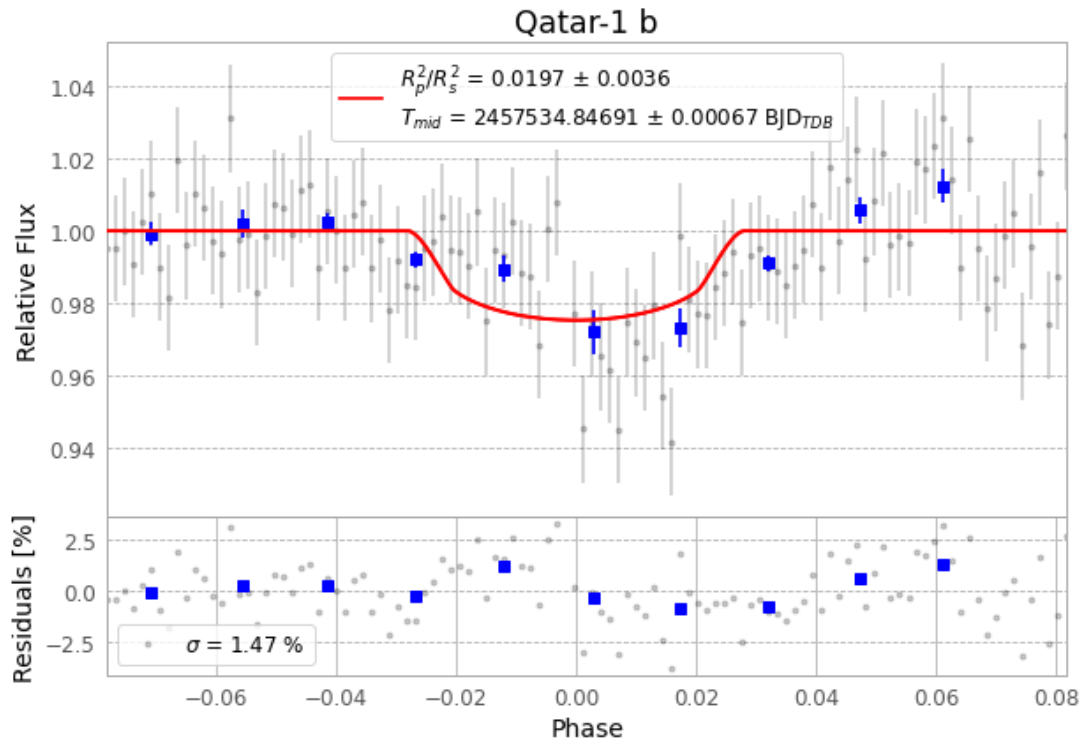


Figure 1 — FinalLightCurve_Qatar-1 b_2016-05-25.png (SHA-256 f3457299913e2ab7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d53657816ba4bbb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

110 FITS frames (72 MB), Qatar-1160526053913.FITS → Qatar-1160526110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-160526.log (d71962bf3ee1...), FinalLightCurve_Qatar-1 b_2016-05-25.png (f3457299913e...), FinalLightCurve_Qatar-1 b_2016-05-25.csv (bb91a2e93973...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 16:09Z.